Practice paper 4 – Exercise 1

Integration and L^p spaces | 7 points | Suggested time: 20–25 minutes

Problem statement. Show your calculations and give brief reasons. For $n \geq 1$, let $f_n(x) = x + 1/n$ on $(0,1)$, with Lebesgue measure, and let $f(x) = x$.

(a) **[2 points]** Show that $f_n(x) \to f(x)$ for every $x \in (0,1)$. Compute $\int_0^1 f_n(x)\,dx$ and its limit.

(b) **[3 points]** Compute $\|f_n - f\|_{L^1}$ and $\|f_n - f\|_{L^2}$. Does $f_n \to f$ in L^1 and in L^2?
Hint: first simplify the difference $f_n(x) - f(x)$.

(c) **[2 points]** Prove $|f_n(x)| \leq 2$ for every $n \geq 1$ and $x \in (0,1)$. Use this integrable bound to apply the dominated convergence theorem and compare your answer with part (a).

Practice paper 4 – Solution 1

Detailed worked solution

(a) Limit and integral. For any fixed $x \in (0,1)$, $1/n \to 0$, so

$$f_n(x) = x + \frac{1}{n} \longrightarrow x = f(x).$$

By linearity of integration,

$$\int_0^1 f_n(x)\,dx = \int_0^1 x\,dx + \frac{1}{n}\int_0^1 1\,dx = \frac{1}{2} + \frac{1}{n} \longrightarrow \frac{1}{2}.$$

Also $\int_0^1 f(x)\,dx = \int_0^1 x\,dx = 1/2$.

(b) Convergence to a nonzero function. To test convergence to f, calculate the norms of the difference, rather than the norms of f_n alone:

$$f_n(x) - f(x) = \left(x + \frac{1}{n}\right) - x = \frac{1}{n}.$$

The difference is constant on an interval of length one. Therefore

$$\|f_n - f\|_{L^1} = \int_0^1 \frac{1}{n}\,dx = \frac{1}{n},$$

$$\|f_n - f\|_{L^2} = \left(\int_0^1 \frac{1}{n^2}\,dx\right)^{1/2} = \sqrt{\frac{1}{n^2}} = \frac{1}{n}.$$

Both norms tend to zero. Thus

$$\boxed{f_n \to f \text{ in } L^1(0,1) \text{ and in } L^2(0,1).}$$

(c) Dominated convergence. Since $0 < x < 1$ and $0 < 1/n \le 1$, we have

$$0 < f_n(x) = x + \frac{1}{n} \le 2.$$

Choose $g(x) = 2$. This function is integrable, since $\int_0^1 2\,dx = 2$. Every f_n is measurable (it is continuous), the pointwise limit is $f(x) = x$, and the bound $|f_n| \le g$ holds for every n. Dominated convergence gives

$$\lim_{n\to\infty} \int_0^1 f_n(x)\,dx = \int_0^1 f(x)\,dx = \boxed{\frac{1}{2}}.$$

This matches part (a).

Key point. Convergence in L^p to a function f means that the norm of $f_n - f$ tends to zero. The norm of f_n need not tend to zero.

Practice paper 4 – Exercise 2

Probability and conditioning | 7 points | Suggested time: 20–25 minutes

Problem statement. Show your calculations and give brief reasons. Let X, Y have joint density

$$f_{X,Y}(x,y) = \begin{cases} 2x, & 0 < x < 1,\ 0 < y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

(a) **[2 points]** Compute the marginal densities of X and Y. Show that X and Y are independent by checking $f_{X,Y} = f_X f_Y$.

(b) **[3 points]** Determine $f_{Y|X=x}$ for $0 < x < 1$, then compute $\mathbb{E}[Y \mid X = x]$ and $\mathbb{E}[Y]$.

(c) **[2 points]** Compute $\mathbb{E}[X]$ and $\mathbb{P}(X < 1/2, Y < 1/2)$.

Hint: for the probability, you may use independence and the marginal densities.

Practice paper 4 – Solution 2

Detailed worked solution

(a) Marginals and independence. On the unit square, integrate with respect to the variable being removed:

$$f_X(x) = \int_0^1 2x\,dy = 2x, \quad 0 < x < 1,$$

$$f_Y(y) = \int_0^1 2x\,dx = \Big[x^2\Big]_0^1 = 1, \quad 0 < y < 1.$$

Both densities are zero outside $(0,1)$, and both integrate to one. On the unit square,

$$f_X(x) f_Y(y) = 2x \cdot 1 = 2x = f_{X,Y}(x,y).$$

Outside it, both sides vanish. Thus X and Y are independent, even though they do not have the same marginal distribution.

(b) Conditional law and expectation. For $0 < x < 1$, we have $f_X(x) > 0$. The conditional density is

$$f_{Y|X=x}(y) = \frac{2x}{2x} = 1, \quad 0 < y < 1,$$

with value zero elsewhere. It is the uniform density on $(0,1)$. Hence

$$\mathbb{E}[Y \mid X = x] = \int_0^1 y\,dy = \left[\frac{y^2}{2}\right]_0^1 = \frac{1}{2}.$$

Thus $\mathbb{E}[Y \mid X] = 1/2$ almost surely, and the tower property gives

$$\mathbb{E}[Y] = \mathbb{E}[\mathbb{E}[Y \mid X]] = \mathbb{E}[1/2] = \boxed{\frac{1}{2}}.$$

(c) Mean of X and joint probability. Using $f_X(x) = 2x$,

$$\mathbb{E}[X] = \int_0^1 x\,2x\,dx = 2\left[\frac{x^3}{3}\right]_0^1 = \boxed{\frac{2}{3}}.$$

The marginal probabilities are

$$\mathbb{P}(X < 1/2) = \int_0^{1/2} 2x\,dx = \frac{1}{4}, \qquad \mathbb{P}(Y < 1/2) = \int_0^{1/2} 1\,dy = \frac{1}{2}.$$

By independence,

$$\boxed{\mathbb{P}(X < 1/2, Y < 1/2) = \frac{1}{4} \cdot \frac{1}{2} = \frac{1}{8}.}$$

Key point. Independence means that the joint law factors into the marginal laws; it does not mean the two marginal laws must coincide.